Horizontal Stabilizer – General Data

1. General

The horizontal stabilizer is housed in a recess in the vertical stabilizer structure, and is secured by four bolts to the rear fuselage structure. A spacing block is inserted between the forward attachments to provide the correct degree of incidence.

The horizontal stabilizer comprises front and rear spars, and ribs reinforced by spanwise stringers. Hinge plates are provided at two places on each side for attachment of elevator hinge brackets, and a central mounting on the rear of the stabilizer supports the elevator torque tubes. As a customer option, de-icer boots may be fitted to the leading edge of the stabilizer.

2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Horizontal Stabilizer

1) Remove vertical stabilizer. (Refer to 55-30-00, Vertical Stabilizer – General Data).

2) Remove elevators. (Refer to 27-30-11, Elevator – Maintenance Practices). Coil and stow loose cables.

3) If de-icer boots are fitted, disconnect pipe lines.

4) Attach sling SD12508 to horizontal stabilizer and take weight. (Refer to 07-00-00, Lifting and Shoring – General Data).

5) Remove four bolts, saddle washers and washers at front and rear attachment fittings of horizontal stabilizer to rear fuselage.

6) Lift horizontal stabilizer clear of aircraft. Remove spacing block, four barrel nuts and sleeves from rear fuselage attachment fittings.

B. Install Horizontal Stabilizer

1) Position spacing block at front attachment fittings on rear fuselage.

2) Attach sling and hoist horizontal stabilizer into position at front and rear attachment fittings on rear fuselage. Ensure spacing block lines up with front attachment fittings.

3) Position four sleeves in fuselage attachment fittings, insert four attachment bolts with saddle washers and washers, and check that plain shank of each bolt is approximately flush with inside surface of each sleeve. If necessary, add washer under existing washer on bolt to achieve this condition.

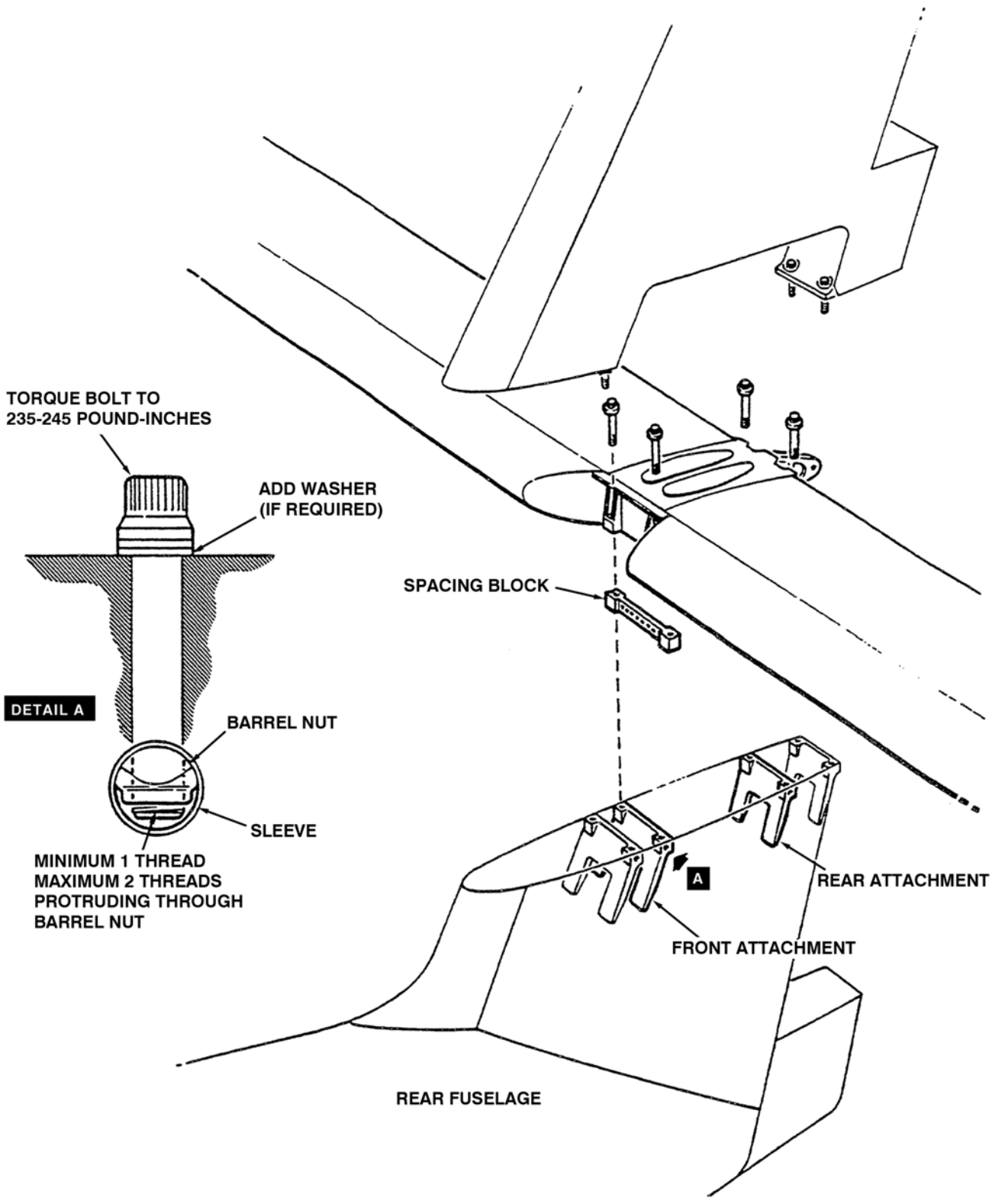
4) Withdraw bolts sufficiently to install barrel nuts in sleeves. Tighten bolts to 235 to 245 pound–inches. Check bolt thread protruding through barrel nut is between one thread minimum and two threads maximum.

5) Remove sling from horizontal stabilizer.

6) If de-icer boots are fitted connect pipe lines.

7) Install elevators. (Refer to 27-30-11, Elevator – Maintenance Practices).

8) Install vertical stabilizer. (Refer to 55-30-00, Vertical Stabilizer – General Data).



Horizontal Stabilizer Installation
Figure 1